

Prak Modul 4

3. - Di Semua Router:

```
cd /usr/lib/frr
```

```
./zebra -d
```

```
./ripd -d
```

```
./mgmtd -d
```

```
vtysh
```

```
conf t
```

```
router rip
```

```
- Router1:
```

```
network 10.28.2.0/25
```

```
network 10.28.2.240/30
```

```
- Router2:
```

```
network 10.28.2.240/30
```

```
network 10.28.2.248/30
```

```
network 10.28.2.244/30
```

```
default-information originate
```

```
- Router3:
```

```
network 10.28.2.244/30
```

```
network 10.28.2.224/28
```

```
network 10.28.2.192/27
```

```
- Router4:
```

```
network 10.28.2.248/30
```

```
network 10.28.2.128/27
```

```
network 10.28.2.160/27
```

```
network 10.28.2.252/30
```

```
- Router5:
```

```
network 10.28.2.252/30
```

```
network 10.28.0.0/23
```

```
- Di Semua Route (Liat kalau ada R di bawah "IPv4 unicast VRF default"):
```

```
Exit
```

```
show ip rip
```

```
show ip route
```

4. – Di kedua web server:

- nano /myscripts/myweb/index.html: [index.html](#)

- nano /myscripts/myconfig/nginx.conf:

```
user www-data;
```

```
worker_processes auto;
```

```
worker_cpu_affinity auto;
```

```
pid /tmp/nginx.pid;
```

```
error_log /myscripts/mylogs/error.log;
```

```
events { worker_connections 768; }
```

```
http {
```

```
    server {
```

```
        listen 80;
```

```
        server_name _;
```

```
        root /myscripts/myweb;
```

```
        index index.html;
```

```
        location / {
```

```
            try_files $uri $uri/ =404;
```

```
        }
```

```
    }
```

```
}
```

- nano /myscripts/myconfig/start_http.sh:

```
#!/bin/bash
```

```
nginx -c /myscripts/myconfig/nginx.conf -g 'daemon off;'
```

- Tes di salah satu host:

```
curl -vL 10.28.2.128:8080 (Web1)
```

```
curl -vL 10.28.2.162:8080 (Web2)
```

5. – Di router-5:

```
iptables -A FORWARD -d 10.28.0.0/23 -p tcp --dport 1337 -j DROP
```

```
iptables -A FORWARD -d 10.28.0.0/23 -p tcp --dport 4444 -j DROP
```

```
iptables-save
```

- Tes di IT-PC:

nc 10.28.0.2 1337

nc 10.28.0.2 4444

- Tes di HR-PC:

nc -l -p 1337

nc -l -p 4444

6. – Di Webserver:

iptables -A INPUT -p tcp --dport 22 -m iprange --src-range 10.28.2.2-10.28.2.77 -j ACCEPT

iptables -A INPUT -p tcp --dport 22 -j DROP

iptables-save

- Tes di IT-PC:

sh root@10.28.2.128

sh [root@10.28.2.162](#)

- Tes di Router5:

sh root@10.28.2.128

sh [root@10.28.2.162](#)

7. – Di DB-Server:

iptables -A INPUT -p tcp -m multiport --dports 80,443 -m time --timestart 07:00 --timestop 22:00 --weekdays Mon,Tue,Wed,Thu,Fri,Sat -j ACCEPT

iptables -A INPUT -j DROP

iptables-save

- Tes di IT-PC:

Hari Benar Jam Benar: date --set="2025-11-21 10:00:00"

Hari Benar Jam Salah: date --set="2025-11-21 05:00:00"

Hari Salah Jam Benar: date --set="2025-11-16 10:00:00"

Hari Salah Jam Salah: date --set="2025-11-16 05:00:00"

telnet 10.28.2.194 80

telnet 10.28.2.194 443

telnet 10.28.2.226 80

telnet 10.28.2.226 443

8. – Di Web Server:

```
iptables -A INPUT -p tcp --dport 80 -j ACCEPT
```

```
iptables -A INPUT -j DROP
```

```
iptables-save
```

- Tes di IT-PC:

```
curl -vL http://10.28.2.128
```

```
curl -vL http://10.28.2.162
```

```
ssh root@10.28.2.128
```

```
ssh root@10.28.2.162
```

9. – Di router-5:

```
iptables -A INPUT -p icmp -d 10.28.0.0/23 -j REJECT
```

```
iptables -A FORWARD -p icmp -d 10.28.0.0/23 -j REJECT
```

```
iptables-save
```

- Tes di IT-PC:

```
ping 10.28.0.2
```

```
ping 10.28.2.2
```

10. – Di Semua Node:

```
apt-get install ulogd2 -y
```

```
service ulogd2 start
```

```
cp /etc/ulogd.conf /root/ulogd.conf
```

```
rm /etc/ulogd.conf
```

```
cp /root/ulogd.conf /etc/ulogd.conf
```

```
service ulogd2 restart
```

- Di Router-5 (untuk testing):

```
iptables -I INPUT 1 -j NFLOG --nflog-prefix "DROPPED PACKET: " --nflog-group 0
```

```
iptables-save
```

- Di Semua Node:

```
iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
```

```
iptables -A INPUT -p tcp --dport 22 -j ACCEPT
```

```
iptables -A INPUT -p icmp -j ACCEPT
```

```
iptables -A INPUT -j NFLOG --nflog-prefix "DROP INPUT: " --nflog-group 0
```

```
iptables -A INPUT -j DROP
```

```
iptables -A FORWARD -j NFLOG --nflog-prefix "DROP FORWARD: " --nflog-group 0
```

```
iptables -A FORWARD -j DROP
```

```
iptables -A OUTPUT -j NFLOG --nflog-prefix "DROP OUTPUT: " --nflog-group 0
```

```
iptables -A OUTPUT -j DROP
```

```
iptables-save
```

- Tes di IT-PC:

```
ssh root@10.28.2.162
```

```
ssh root@10.28.2.128
```

```
ping 10.28.0.2
```

```
ping 10.28.2.2
```

- Cek Log di Router-5:

```
cat var/log/ulog/syslogemu.log
```